Fair AI in cloud based music streaming

D4F Assignment

Use case: Fair AI in cloud based music streaming

Scenario
Ling is a passionate music lover. She has been listening to music for her whole life. As her music streaming platform she currently uses spotify. She has been using spotify for years so she had build up a big data profile including favorite songs, playlists and a perfect recommendation algorithm. Because Spotify is too expensive for Ling and Ling wants enhanced access to exclusive content, she decides to switch to another music streaming platform. Upon switching, Ling encounters a big problem. Her playlists, favorite songs and personalized recommendations cannot be automatically transferred to her new music streaming platform. This means she has to start over creating a new list of favorite songs, playlists and her recommendations will not be that good because there is no data to base it on. Manually re-creating these playlists and finding specific albums again is extremely time consuming and frustrating (Gao, 2023).

Main tasks and user needs:

- Ling needs to access her existing playlists and favorite tracks easily.
- Ling needs to export her music library. Including playlists, favorite songs and other metadata used for the recommendation algorithm from the old streaming service.
- Ling needs this data to be imported in the new platform in the most usable way.
- Ling needs to receive a notification if certain albums or songs are unavailable on the new platform.

Context/Environment of the system
The scenario involves cloud-based music streaming platforms like Spotify, Apple Music, Tidal, or Deezer, where user data, playlists, and music libraries are stored remotely within the cloud.

Current situation and future vision
Currently, music streaming platforms lack standardized methods for exporting and importing user data between different services, causing inconvenience and limiting user flexibility. It is also bad for the equality and diversity because new users at the music platform are getting less good and diverse recommendations because there is no data to base the recommendations on yet. The two current options for data transferring is manual, which is very unusable and through a third party which is bad for your data privacy because you share your data with the third party (Chaabane, Kaafar, & Boreli, 2013).

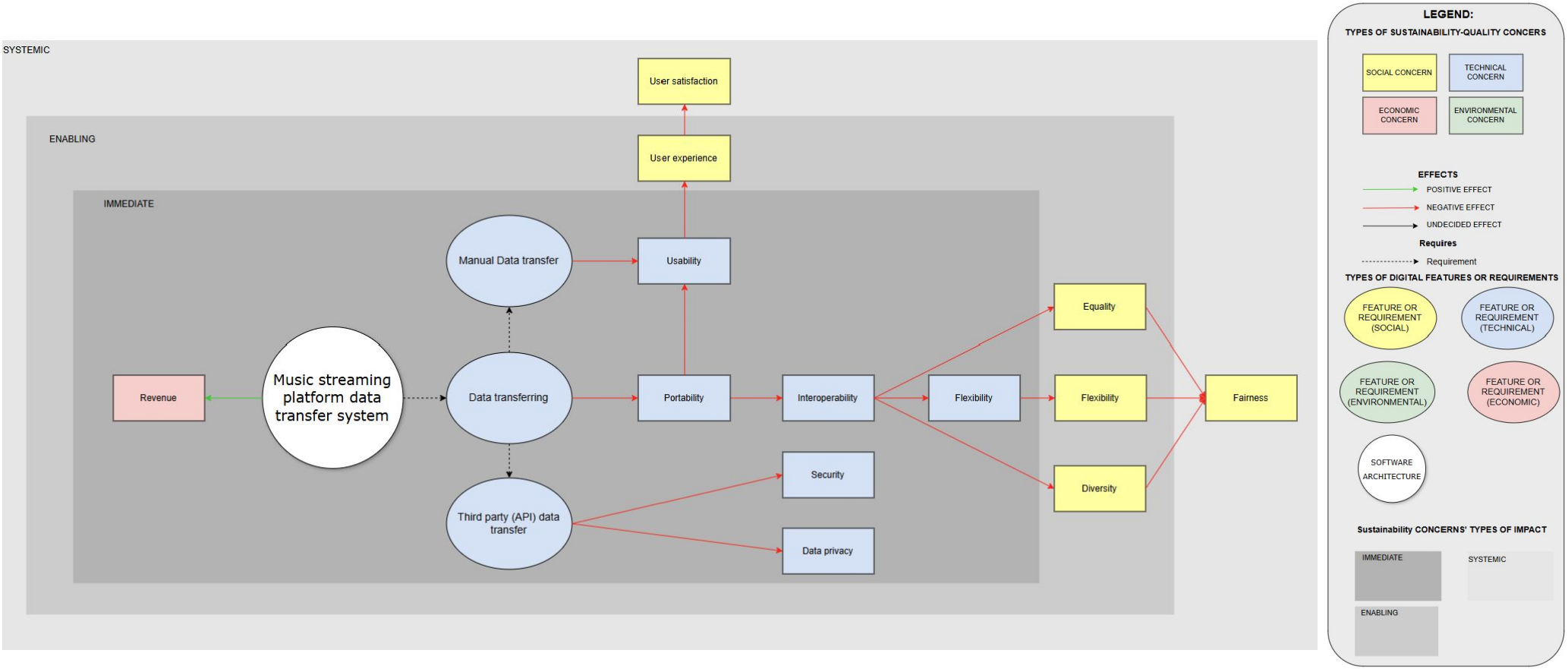
The envisioned future system would create an API integration for all possible data transfers between different music streaming platforms.

Design challenges

- Enabling compatibility, interoperability and standardization across various streaming services for data export/import.
- Efficiently transferring user listeners data between platforms (playlists, favorites).
- Handling missing albums or songs in the new platform.

Why is this system needed?
This system is essential to improve usability, flexibility, interoperability, user satisfaction and overall fairness of the system. It facilitates smooth transitions between streaming services, preserving personalized music experiences and minimizing frustration and manual effort for users like Ling.

Definition of Fairness: Decision Map



Definition of Fairness: Decision Map (cont.)

The main elements of our decision map to be considered are that the platforms currently create revenue for themselves by not allowing users to be the owner of their own data by making it less attractive to switch platforms (Gao, 2023).

The core issue lies in the lack of a music streaming data transfer system. In the current setup, manual transfer is not usable and third-party APIs raise privacy and security concerns (Chaabane, Kaafar, & Boreli, 2013). The current data transfer requirement has an bad effect on portability, usability, and interoperability.

Without these, user experience suffers, flexibility to switch platforms. As new users receive less personalized recommendations. This hurts diversity and equality, which impacts fairness.

To get more information about the definitions of the elements the paper of (Lago, Patricia. 2019) could be consulted

Definition of Fairness: SQ model

FILL IN YOUR OWN QA NAME	FILL IN YOUR OWN QA DEFINITION	SUSTAINABILITY DIMENSION OF THE QA				FILL IN MEASURES IN USE (if applicable)
QUALITY ATTRIBUTE (QA)	DEFINITION	TECHNICAL	ENVIRONMENTAL	ECONOMIC	SOCIAL	DEFINITION of METRIC/KPI
Revenue	The financial impact on music streaming platforms. It includes potential revenue loss from users switching to another streaming platform.			X		Subscription price (€) per customer # of paying customers
Flexibility	The (technical) ability for users to switch easily between music streaming platforms without losing their data, playlists and personalized recommendations.	X			X	
Equality	Ensuring all users within a music streaming platdorm, whether new or existing, receive a similar level of music personalization recommendations preventing biases against newcomers who have lack of historical data.				X	% of satisfactory recommendations of new users % of satisfactory recommendations for existing users
Portability	The ability to seamingly transfer data in the same format between different applications in the same of different architectures	X				
Diversity	The ability of the recommendation algorithm within a music streaming platform to continue offering a diverse selection of music. Also for new users on the platform without having much personal data.				X	
Data privacy	Protecting user data during export/import processes and ensuring that there is no third-party data access needed. This could expose private information.	X				
Security	Ensuring that data transfers between platforms are encrypted and protected against breaches or unauthorized access.	X				
Interoperability	The ability of different music streaming platforms to communicate and exchange data with each other.	X				# of platforms offering the exchance of data
Usability	Designing the system in the way it is easy to understand and perform tasks for the users	X				Time (in seconds) it takes for users understand all basic functionalities
User experience	The way users are experience the use of the system. In this case especially the data tranfer process.				X	% of customers using the transfer
User satisfaction	The overall way how users are feeling with the data transfer process.				X	% of satisfied customers

Definition of Fairness: SQ model (cont.)

The SQ models shows our quality attributes with their definition regarding the Music Streaming Services and what quality attributes are impacted by the unfairness of the current situation regarding cloud storage. It strongly impacts interoperability, usability and the user experience for end-users, who are no longer the owner of their own data.

Fairness perspective

	Description
Desired quality	The ability for users like ling to easily and securely transfer music cloud-data in a usable way to switch between music platforms, to receive a similar and fair music experience without third party data-access.
Concerns	The fairness definition adresses the following concerns: Portability, Interoperability, Flexibility, Diversity and Equality, which are all negatively impacted by the current state-of-start of cloud-stored data transfer between music streaming platforms.
Architectural tactic(s)	Standardized Music Streaming platform data portability
Possible KPI(s)	Sustainable data access and transfer between music streaming platforms.
Problems and pitfalls	Risk of resistance from streaming providers due to possible loss in revenue, risk of data privacy issue when regulations are not enforced.

made according to: I. Fatima, M. Funke and P. Lago, "Providing Guidance to Software Practitioners: A Framework for Creating KPIs," in *IEEE Software*, doi: 10.1109/MS.2024.3456446. keywords: {Key performance indicator;Organizations;Software measurement;Unified modeling language;Measurement;Interviews;Monitoring},

Definition of KPI

Sustainable data access and transfer between music streaming platforms

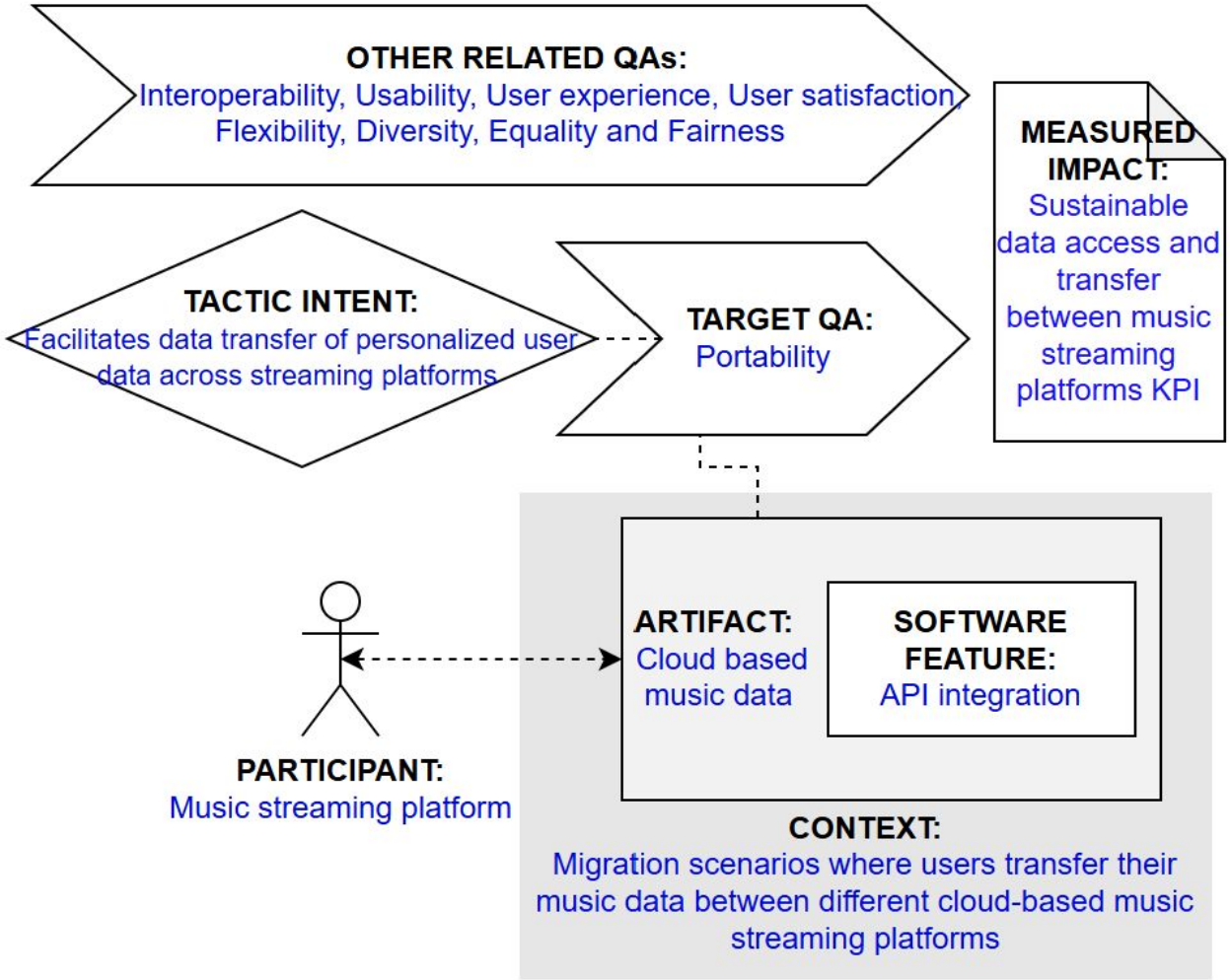
Goal	Critical Success Factor (CSF)	Key Performance Indicator (KPI)		Metrics*	Measures
To let end-users of music streaming services easily switch between services, without losing their data (e.g. playlist, liked songs, metadata, etc.).	By 2028, end-users of music streaming services should be the owner of their own cloud-stored data, having access to an interface that provides easy data transfer (export and import) between streaming interfaces using a standardized API.	The percentage (%) of music streaming end-users* having access to a clear interface to transfer cloud-stored user-data to other music streaming providers: satisfied users (su) = users * (usability + completeness + availability / 3) **% = su / users * 100** *we use end-users instead of provides, to weigh providers with more users heavier		**Usability** *How easy to use is the process of data transfering* usability = SUS score / 4 / ceil(transfer time) [1] **Completeness** *How complete is transferred data (how much data loss)* completeness = (exported data + imported data) / 2 / 100 **Availability** *How available is the transfer functionality to end-users* availability = ((API uptime / 100) + (providing services / services)) / 2 *all metrics have a maximum value of 1	**Users:** total amount of end users of music streaming services (#) **System Usability Score (SUS) score [1]:** average score given by users *(0-4)* **Transfer time:** Time to complete transfer after process started *(s)* **Exported data:** Cloud-stored data able to export *(%)* **Imported data:** Cloud-stored data able to import *(%)* **API uptime** *(%)* **Providing services :** Streaming services providing a transferring interface *(#)* **Services:** all streaming services *(#)*
		Target	**Actions**		
		Percentage of end-users having access to easy cloud-stored data transfer > **80%**	Create an ontology for standardized data transfer, create an implementation framework so providers can easily implement it into their application, perform analysis to understand why streaming services are not implementing it.		

[1] Kaya, A., Ozturk, R., Altin Gumussoy, C. (2019). Usability Measurement of Mobile Applications with System Usability Scale (SUS). In: Calisir, F., Cevikcan, E., Camgoz Akdag, H. (eds) Industrial Engineering in the Big Data Era. Lecture Notes in Management and Industrial Engineering. Springer, Cham. https://doi.org/10.1007/978-3-030-03317-0_32

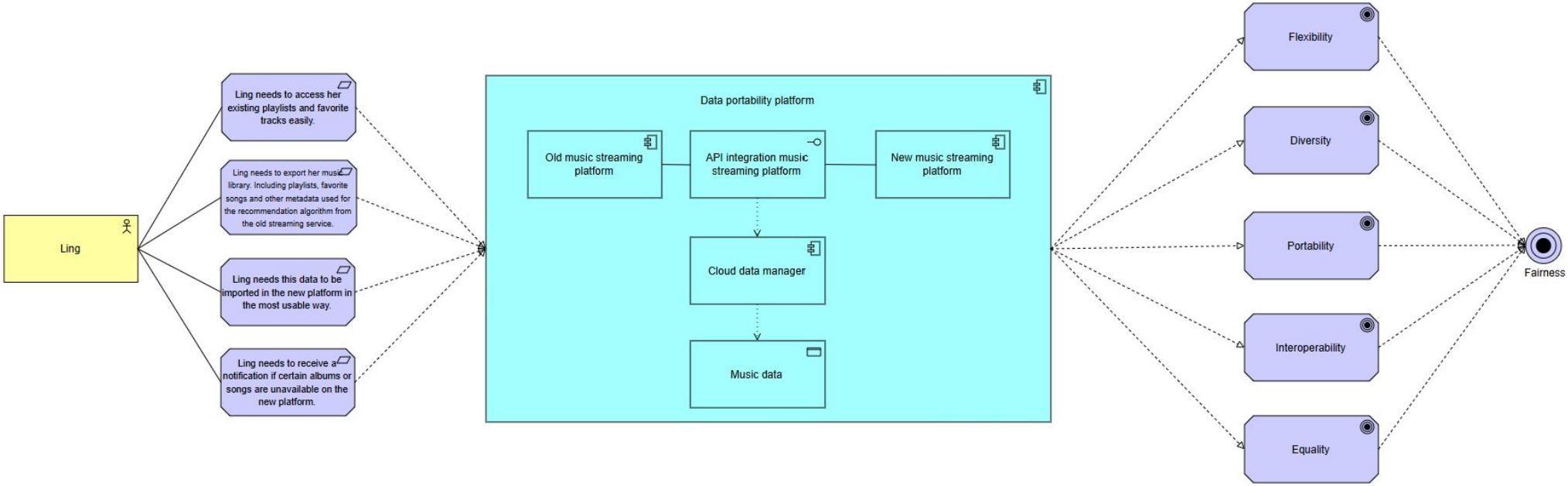
Architectural Tactic for Fairness

Attribute	Description
Title	Standardized Music Streaming Platform Data Portability
Tactic Sort	Awesome tactic
Description	Establish standardized APIs and data schemas across music streaming platforms, enabling automated, secure, and usable transfer of playlists, favorite tracks, and recommendation metadata when users want to use another streaming platform.
Participant	Music streaming platforms
Artifact	Cloud-based music streaming platform data. This includes playlists, favorite tracks, and recommendation metadata
Context	Data transfer scenarios where the users wants to transfer their music data between different cloud-based streaming platforms
Feature	Standardized data export/import API integration
Intent	Facilitates data transfer of personalized user data across cloud based music streaming platforms to get better fairness and user satisfaction.
Target QA	Portability
Related QAs	Interoperability, Usability, User Experience, User Satisfaction, Flexibility, Diversity, Equality, Fairness
Measured Impact	Measures from KPI Sustainable data access and transfer between music streaming platforms
Source	Data Portability among Online Platforms (Engels, Barbara. 2016)
Source DOI	10.14763/2016.2.408

Architectural Tactic for Fairness model



Fairness view: Data portability platform view



The Open Group. (2016). ArchiMate® 3.0 specification (Document Number: C162). The Open Group. https://pubs.opengroup.org/architecture/archimate3-doc/index.html

Fairness view (cont.)

To make the fairness view about this use case we have used archimate to provide a clear model. More information about the notation can be found in in the archimate specification. (The Open Group. 2016)

First, on the left side, the requirements of the business actor, Ling are showed. This are the base requirements that will be needed to receive fairness. To achieve the requirements we need a data portability platform which, as shown in the model. The platform connects the old music streaming platform to the new one through an API integration. A cloud data manager handles the flow of music data. This enables a structured and secure transfer between services. This platform directly addresses Ling's needs by automating the export, import, and compatibility of her data.

By using the new platform, it supports capabilities such as portability, interoperability, flexibility, diversity and equality. Ling can move her data, so this gives portability (Engels, Barbara. 2016). The integration between the music streaming platforms ensures interoperability. The freedom to switch platforms without losing value gives Ling flexibility and the maintaining of personalized content and recommendations brings Ling equality and diversity.

Together, these capabilities lead to fairness. This makes sure users like Ling are not penalized or for switching platforms or prevented to switch. This ensures that all users, new or returning, receive equal access to high-quality, personalized experiences. The model emphasizes that fairness in digital services starts with giving users control over their own data.

The Open Group. (2016). ArchiMate® 3.0 specification (Document Number: C162). The Open Group. https://pubs.opengroup.org/architecture/archimate3-doc/index.html

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